Skills Summary

Substitution: the Chain Rule, Read Backwards

Use this reference while completing your worksheet or when studying.

Skill 1 — Spotting the Inner Function

Look for a pair. An inner function and a constant multiple of its derivative, both in the same integrand. The inner function becomes u.

Build the three-row table every time: $u = \ldots$, then $du = \ldots\, dx$, then the integral rewritten with no x left in it. If an x survives, u was the wrong choice.

Example: Evaluate $\displaystyle\int 3x^2\left(x^3+2\right)^4 dx$.

Step 1. The bracket is an inner function and the factor in front is exactly its derivative, which is the signal to substitute rather than expand.

Step 2. $u = x^3+2$, $du = 3x^2\,dx$, so the integral is $\int u^4\,du = \frac{u^5}{5} + C$.

$\Rightarrow \quad \boldsymbol{\frac{(x^3+2)^5}{5} + C}$

Try it: Evaluate $\displaystyle\int 4x^3\left(x^4+1\right)^2 dx$.

check: $\frac{(x^4+1)^3}{3} + C$

Skill 2 — Indefinite: Convert Back to x

Missing constant? If du is a multiple of what you have, solve for it: $du = 2x\,dx$ means $x\,dx = \frac{1}{2}du$, and that $\frac{1}{2}$ rides outside the integral.

An indefinite answer must come back to x and carry $+C$. Differentiate it mentally as a check — the chain rule should hand back the integrand.

Example: Evaluate $\displaystyle\int x\sqrt{x^2+4}\,dx$.

Step 1. The radicand is the inner function and x is half its derivative, so substitute and carry the missing one half outside.

Step 2. $u = x^2+4$, $x\,dx = \frac{1}{2}du$, so $\frac{1}{2}\int u^{1/2}du = \frac{1}{3}u^{3/2} + C$.

$\Rightarrow \quad \boldsymbol{\frac{(x^2+4)^{3/2}}{3} + C}$

Try it: Evaluate $\displaystyle\int x\sqrt{x^2+1}\,dx$.

check: $\frac{(x^2+1)^{3/2}}{3} + C$

Skill 3 — Definite: Convert the Limits

Rule. In a definite integral, move the limits into the new variable at the same moment the integrand moves:

$$\int_a^b f(g(x))g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du.$$

Then evaluate in u and stop — there is nothing to convert back, and no $+C$.

Example: Evaluate $\int_0^1 3x^2\left(x^3+1\right)^2 dx$.

Step 1. This is a definite integral, so converting the limits is faster and safer than converting the antiderivative back to x.

Step 2. $u = x^3+1$ runs from 1 to 2, so the value is $\left[\frac{u^3}{3}\right]_1^2 = \frac{8-1}{3}$.

$\Rightarrow$ $\boldsymbol{\frac{7}{3}}$

Try it: Evaluate $\int_0^2 2x\left(x^2+1\right)^2 dx$.

check: $\boldsymbol{\frac{124}{3}}$